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11.7.3 Direction of Translocation

Phloem translocates the products of photosynthesis from leaves to the area of growth and storage, in the following directions,

Downward direction: From leaves to stem and roots.

Upward direction: From leaves to developing buds, flowers, fruits for consumption and storage. Germination of seeds is also a good example of upward translocation.

Radial direction: From cells of pith to cortex and epidermis, the food materials are radially translocated.

11.7.4 Source and Sink

Source is defined as any organ in plants which are capable of exporting food materials to the areas of metabolism or to the areas of storage. Examples: Mature leaves, germinating seeds.

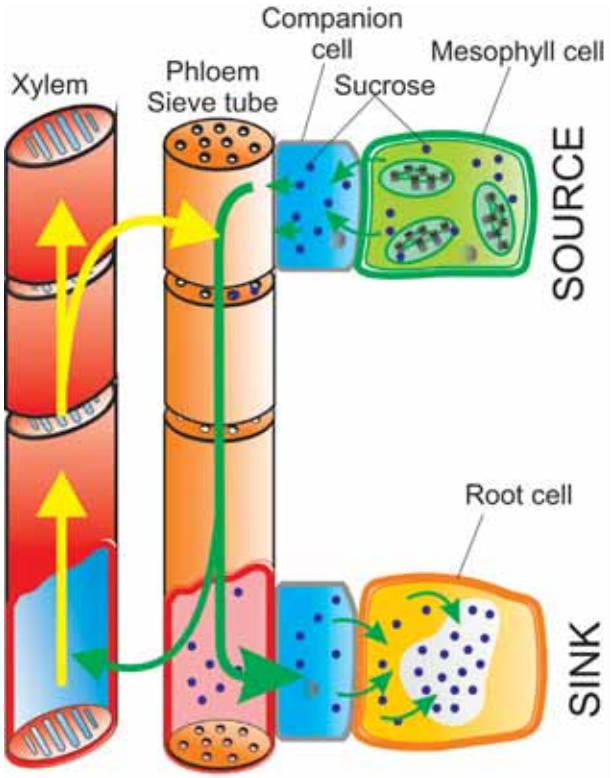


Figure 11.21: Source and Sink

Sink is defined as any organ in plants which receives food from source.Example: Roots, tubers, developing fruits and immature leaves (Figure 11.21).

11.7.5 Phloem Loading

The movement of ***photosynthates*** (products of photosynthesis) from mesophyll cells to phloem sieve elements of mature leaves is known as **phloem loading**. It consists of three steps.



Why plants transport sugars as sucrose and not as starch or glucose or fructose?

Glucose and Fructose are simple monosaccharides, whereas, Sucrose is a disaccharide composed of glucose and fructose. Starch is a polysaccharide of glucose. Sucrose and starch are more efficient in energy storage when compared to glucose and fructose, but starch is insoluble in water. So it cannot be transported via phloem and the next choice is sucrose, being water soluble and energy efficient, sucrose is chosen as the carrier of energy from leaves to different parts of the plant. Sucrose has low viscosity even at high concentrations and has no reducing ends which makes it inert than glucose or fructose.During photosynthesis, starch is synthesized and stored in the chloroplast stroma and sucrose is synthesized in the leaf cytosol from which it diffuses to the rest of the plant.